

WIPRO CAPSTONE PROJECT MODULE 3

AI/ML-Based Predictive Maintenance and Vehicle Health Monitoring System

Submitted By

Parthib Ranjan Sarbajna

Enrollment Number: 12023002028150

Registration Number: 304202300901068

Institute of Engineering and Management, Kolkata

B.Tech in CSE(AI & ML)

Project Report

2026

Abstract

Predictive maintenance is an important application of Artificial Intelligence and Machine Learning in the automotive domain. Instead of relying entirely on fixed maintenance schedules or waiting for component failures, predictive maintenance systems analyse sensor measurements to estimate the likelihood of a fault and provide an early indication of vehicle health.

This project presents an AI/ML-based Predictive Maintenance and Vehicle Health Monitoring System developed as part of the Wipro Capstone Project Module 3. The system uses real engine sensor data obtained from the Hugging Face dataset `jkskswamy/predictive-maintenance`. The dataset contains six sensor parameters: Engine RPM, Lub Oil Pressure, Fuel Pressure, Coolant Pressure, Lub Oil Temperature, and Coolant Temperature. The target variable, Engine Condition, represents whether the engine is operating under a normal condition or requires maintenance.

The project implements data preprocessing, exploratory data analysis, machine learning model development, model comparison, feature importance analysis, SHAP-based explainability, and vehicle health monitoring. Two tree-based classification models, Random Forest and XGBoost, were trained and evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

The Random Forest model achieved a ROC-AUC of approximately 68.83%, while XGBoost achieved approximately 68.60%. Since ROC-AUC was used as the model-selection criterion, the Random Forest model was selected for the vehicle health monitoring stage.

The final system converts the predicted failure probability into a vehicle health score ranging from 0 to 100. The system also categorises the vehicle condition into NORMAL, MONITOR, WARNING, or CRITICAL states. SHAP analysis was used to improve interpretability and identify the relative contribution of sensor parameters to model predictions.

The project demonstrates how machine learning and explainable AI techniques can be combined to create a practical predictive maintenance framework using real sensor data.

Contents

Abstract	i
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Project Objective	2
1.4 System Architecture	2
1.5 Scope of the Project	3
2 Technical Background	4
2.1 Machine Learning for Predictive Maintenance	4
2.2 Classification	4
2.3 Random Forest	4
2.4 XGBoost	5
2.5 SHAP	5
3 Dataset and Data Analysis	6
3.1 Dataset Source	6
3.2 Dataset Features	6
3.3 Target Variable	6
3.4 Data Preprocessing	7
3.5 Engine Condition Distribution	7
3.6 Sensor Distributions	8
3.7 Correlation Analysis	9
4 Methodology	10
4.1 Overall Methodology	10
4.2 Train-Test Split	10
4.3 Random Forest Model	11
4.4 XGBoost Model	11
4.5 Model Comparison	11

4.6	ROC Curve	12
4.7	Confusion Matrix	12
5	Feature Importance and Explainability	14
5.1	Random Forest Feature Importance	14
5.2	XGBoost Feature Importance	14
5.3	SHAP Explainability	15
5.4	SHAP Feature Importance	15
6	Vehicle Health Monitoring	17
6.1	Failure Probability	17
6.2	Vehicle Health Score	17
6.3	Health Status Categories	18
6.4	Vehicle Health Score Visualisation	18
6.5	Example Prediction	18
7	Experimental Setup	20
7.1	Software Environment	20
7.2	Machine Learning Models	20
7.3	Evaluation Metrics	21
7.3.1	Accuracy	21
7.3.2	Precision	21
7.3.3	Recall	21
7.3.4	F1 Score	21
7.3.5	ROC-AUC	21
8	Results and Analysis	22
8.1	Model Performance	22
8.2	Model Selection	22
8.3	Feature Importance Results	22
8.4	XGBoost Feature Importance Values	23
8.5	Overall Result	23
9	Discussion	24
9.1	Model Behaviour	24
9.2	Importance of Engine RPM	24
9.3	Explainability	24
9.4	Health Monitoring	25
10	Limitations	26

11 Future Scope	27
11.1 Real-Time Sensor Integration	27
11.2 Additional Sensor Parameters	27
11.3 Deep Learning	27
11.4 Remaining Useful Life Prediction	28
11.5 Real-Time Dashboard	28
11.6 Automated Maintenance Alerts	28
12 Conclusion	29
A Project Directory Structure	31
B Health Score Calculation	33
C Repository Information	34

List of Figures

3.1	Distribution of Engine Condition Classes	8
3.2	Distribution of Engine Sensor Parameters	8
3.3	Correlation Matrix of Engine Sensor Parameters	9
4.1	Comparison of Random Forest and XGBoost Performance	11
4.2	ROC Curves of the Classification Models	12
4.3	Confusion Matrix of the Random Forest Model	13
5.1	Random Forest Feature Importance	14
5.2	XGBoost Feature Importance	15
5.3	SHAP Summary Plot	15
5.4	SHAP-Based Feature Importance	16
6.1	Vehicle Health Score	18

List of Tables

3.1	Sensor Features Used in the Project	6
3.2	Engine Condition Labels	7
5.1	SHAP Feature Importance	16
6.1	Vehicle Health Status Thresholds	18
6.2	Representative Vehicle Health Prediction	18
7.1	Software Libraries Used	20
8.1	Model Performance Comparison	22
8.2	XGBoost Feature Importance	23

Chapter 1

Introduction

1.1 Background

Modern vehicles contain numerous sensors that continuously monitor engine and component operating conditions. These measurements can be used to identify abnormal operating behaviour and provide information about the health of the vehicle.

Traditional maintenance approaches generally depend on fixed service intervals or reactive maintenance after a component has already failed. Such approaches can result in unnecessary maintenance or unexpected breakdowns.

Predictive maintenance attempts to address this problem by using historical and real-time sensor measurements together with machine learning models. The model learns relationships between sensor measurements and previously observed engine conditions and then estimates the probability that a new observation corresponds to a maintenance-required condition.

1.2 Problem Statement

The objective of this project is to develop a machine-learning-based system capable of predicting engine condition from sensor measurements and converting the prediction into an interpretable vehicle health score.

The system should:

- Process real engine sensor data.
- Perform appropriate data cleaning and preprocessing.
- Analyse relationships among sensor parameters.
- Train machine learning classification models.
- Compare model performance using standard evaluation metrics.

- Identify important sensor parameters.
- Provide explainability using SHAP.
- Estimate failure probability.
- Convert failure probability into a vehicle health score.
- Categorise the predicted vehicle condition.

1.3 Project Objective

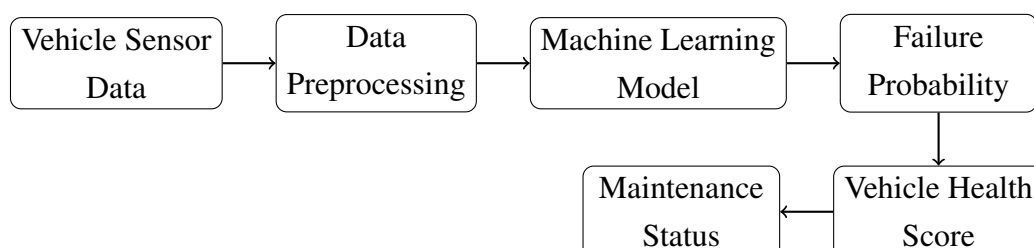
The primary objective is to develop an AI/ML-based predictive maintenance framework that can estimate engine condition from sensor data and provide an understandable indication of vehicle health.

The specific objectives are:

1. Obtain and analyse real engine sensor data.
2. Prepare the dataset for machine learning.
3. Perform exploratory data analysis.
4. Train Random Forest and XGBoost classification models.
5. Compare model performance.
6. Analyse feature importance.
7. Apply SHAP for model explainability.
8. Calculate failure probability.
9. Generate a vehicle health score between 0 and 100.
10. Categorise vehicle health into meaningful maintenance states.

1.4 System Architecture

The overall system follows the pipeline:



1.5 Scope of the Project

The project focuses on predictive engine condition classification using six sensor parameters available in the selected real-world dataset.

The implemented features are:

- Engine RPM
- Lub Oil Pressure
- Fuel Pressure
- Coolant Pressure
- Lub Oil Temperature
- Coolant Temperature

The project does not fabricate or infer unavailable parameters such as vibration, battery voltage, vehicle speed, or operating hours because these parameters are not present in the selected dataset.

Chapter 2

Technical Background

2.1 Machine Learning for Predictive Maintenance

Machine learning allows a computer system to learn patterns from historical data and use those patterns to make predictions on previously unseen observations.

In predictive maintenance, historical sensor measurements are associated with known equipment conditions. A supervised machine learning model can learn the relationship between the sensor values and the corresponding maintenance condition.

2.2 Classification

This project treats engine condition prediction as a binary classification problem.

The target variable is:

$$Y \in \{0, 1\} \tag{2.1}$$

where:

- $Y = 0$: Normal / Healthy condition
- $Y = 1$: Maintenance Required / Faulty condition

2.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees.

Each tree produces a prediction, and the final classification is obtained from the combined predictions of the individual trees.

Random Forest is useful for tabular sensor data because it can model nonlinear relationships and interactions among features.

2.4 XGBoost

XGBoost is a gradient boosting algorithm based on an ensemble of decision trees. Trees are constructed sequentially, with later trees attempting to correct errors made by earlier trees.

XGBoost is widely used for structured and tabular datasets and can capture nonlinear relationships between sensor parameters and the target variable.

2.5 SHAP

SHAP, or SHapley Additive exPlanations, is an explainability method used to estimate how individual features contribute to a model prediction.

For a given prediction, SHAP values indicate whether a feature contributes towards or away from the predicted class.

This allows the predictive maintenance system to provide more interpretable results instead of only returning a classification.

Chapter 3

Dataset and Data Analysis

3.1 Dataset Source

The project uses the publicly available predictive maintenance dataset:

```
jskswamy/predictive-maintenance-data
```

The dataset was accessed using the Hugging Face Datasets library.

3.2 Dataset Features

The six sensor parameters used in the project are shown in Table 3.1.

Table 3.1: Sensor Features Used in the Project

Feature	Description
Engine RPM	Engine rotational speed
Lub Oil Pressure	Lubricating oil pressure
Fuel Pressure	Fuel system pressure
Coolant Pressure	Cooling system pressure
Lub Oil Temp	Lubricating oil temperature
Coolant Temp	Coolant temperature

3.3 Target Variable

The target variable is `Engine Condition`.

Table 3.2: Engine Condition Labels

Value	Meaning
0	Normal / Healthy
1	Maintenance Required / Faulty

3.4 Data Preprocessing

The dataset was processed before model training.

The preprocessing workflow consisted of:

1. Loading the dataset.
2. Selecting the relevant sensor features.
3. Identifying missing values.
4. Removing or handling invalid observations.
5. Separating input features from the target variable.
6. Splitting the data into training and testing subsets.

3.5 Engine Condition Distribution

The distribution of the target classes was visualised to understand the classification problem.

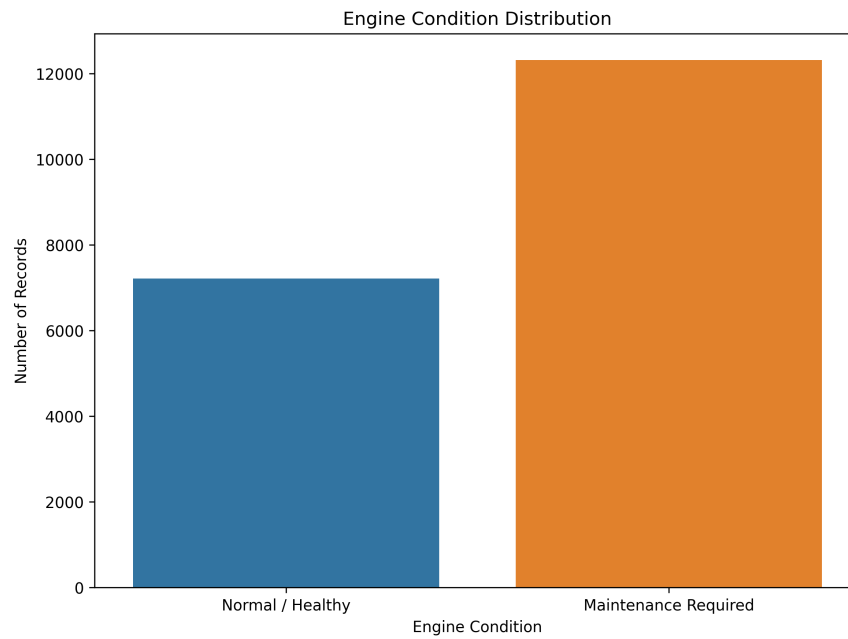


Figure 3.1: Distribution of Engine Condition Classes

Figure 3.1 shows the distribution of normal and maintenance-required engine observations.

3.6 Sensor Distributions

The distributions of the six sensor parameters were analysed before model training.

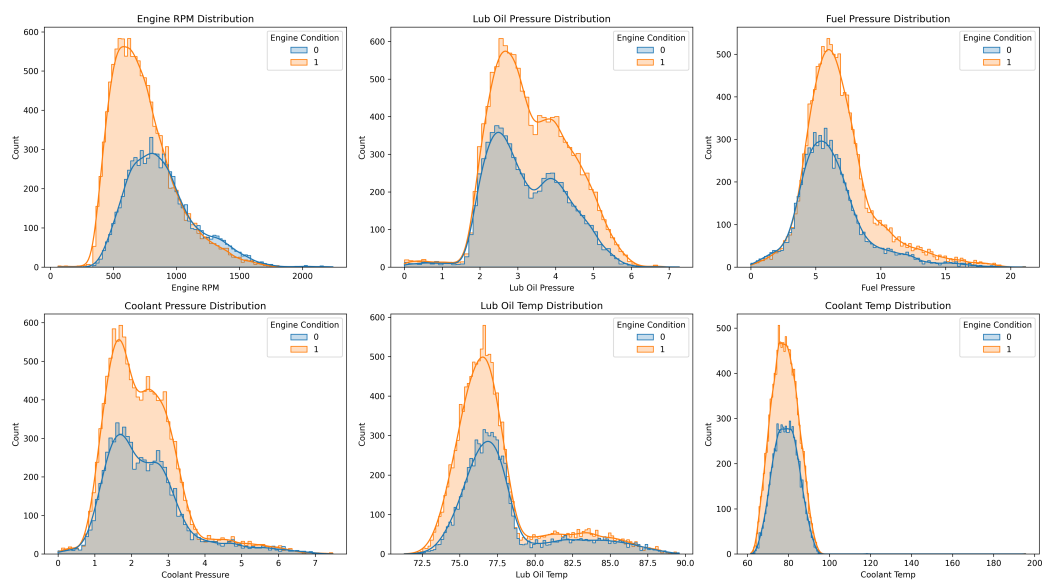


Figure 3.2: Distribution of Engine Sensor Parameters

The distributions provide an overview of the range and variation present in the sensor measurements.

3.7 Correlation Analysis

Correlation analysis was performed to understand relationships among the sensor parameters.

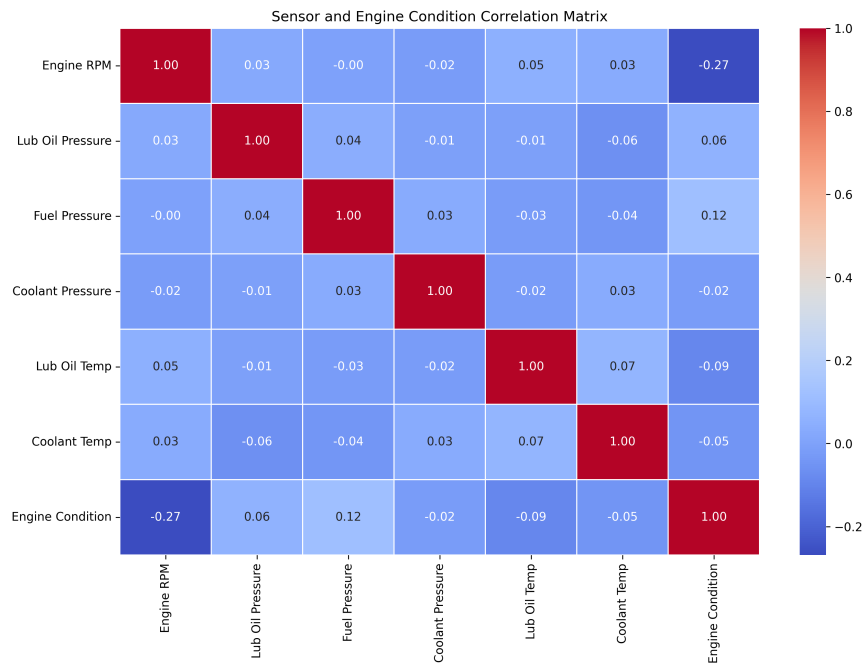


Figure 3.3: Correlation Matrix of Engine Sensor Parameters

The correlation matrix provides a visual representation of relationships among the six sensor variables.

Chapter 4

Methodology

4.1 Overall Methodology

The complete machine learning workflow consists of the following stages:

1. Dataset acquisition
2. Data cleaning
3. Exploratory data analysis
4. Feature-target separation
5. Train-test split
6. Random Forest training
7. XGBoost training
8. Model evaluation
9. Feature importance analysis
10. SHAP explainability
11. Failure probability estimation
12. Vehicle health score generation

4.2 Train-Test Split

The processed dataset was divided into training and testing subsets. The training data was used to fit the machine learning models, while the test data was used to evaluate their performance on unseen observations.

4.3 Random Forest Model

A Random Forest classifier was trained using the six sensor parameters.

The model generates a probability for each class. For predictive maintenance, the probability associated with the maintenance-required class is used as the estimated failure probability.

4.4 XGBoost Model

An XGBoost classifier was also trained using the same input features and target variable.

The use of the same feature set allows a direct comparison between the two models.

4.5 Model Comparison

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

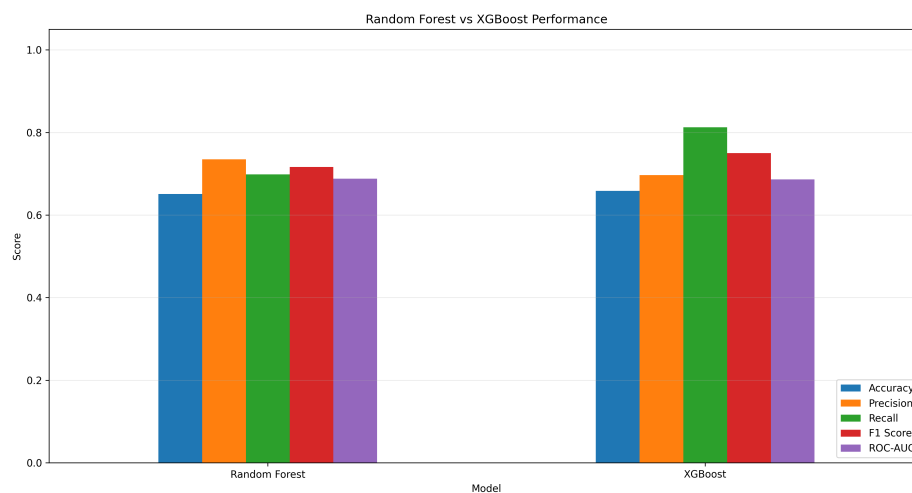


Figure 4.1: Comparison of Random Forest and XGBoost Performance

4.6 ROC Curve

ROC analysis was performed to compare the classification behaviour of the models across different classification thresholds.

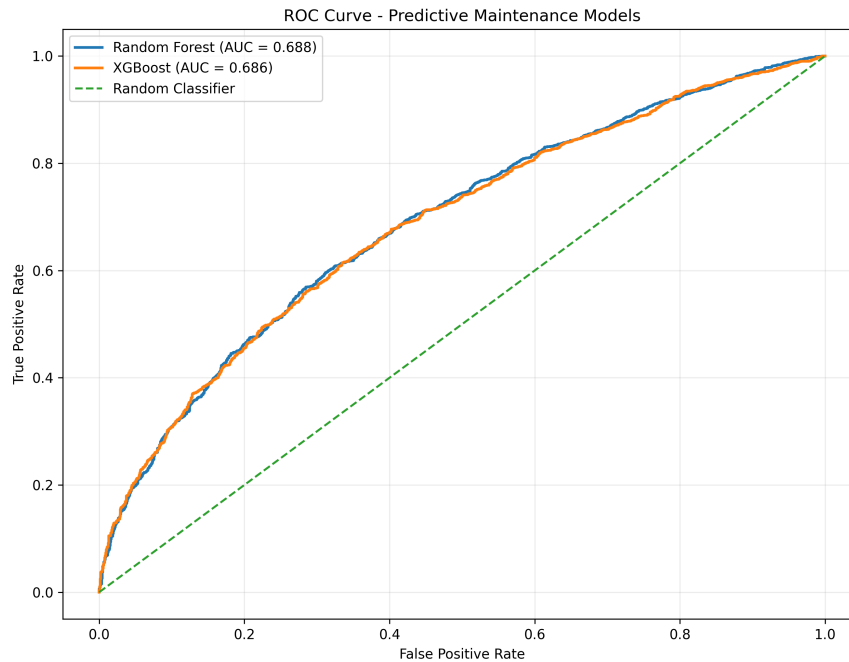


Figure 4.2: ROC Curves of the Classification Models

4.7 Confusion Matrix

A confusion matrix was generated to examine the classification results of the selected Random Forest model.

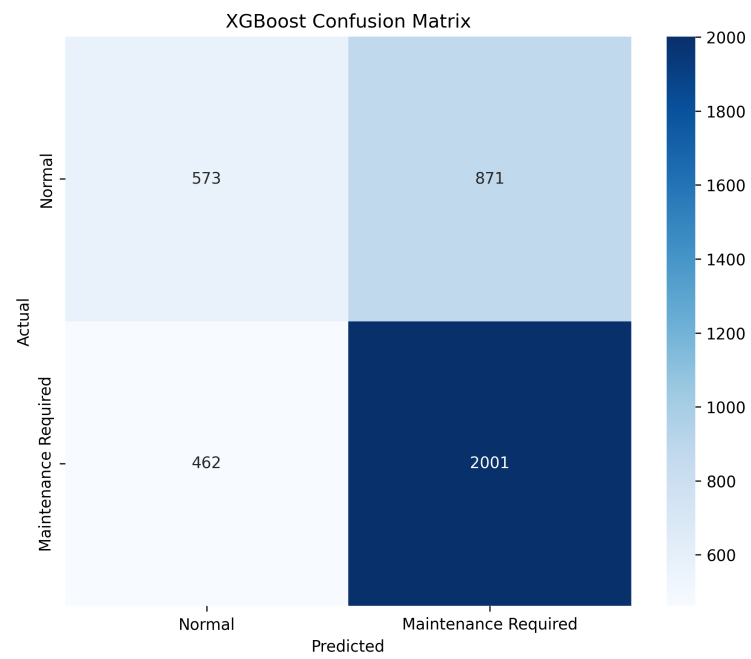


Figure 4.3: Confusion Matrix of the Random Forest Model

Chapter 5

Feature Importance and Explainability

5.1 Random Forest Feature Importance

Feature importance was calculated using the Random Forest model.

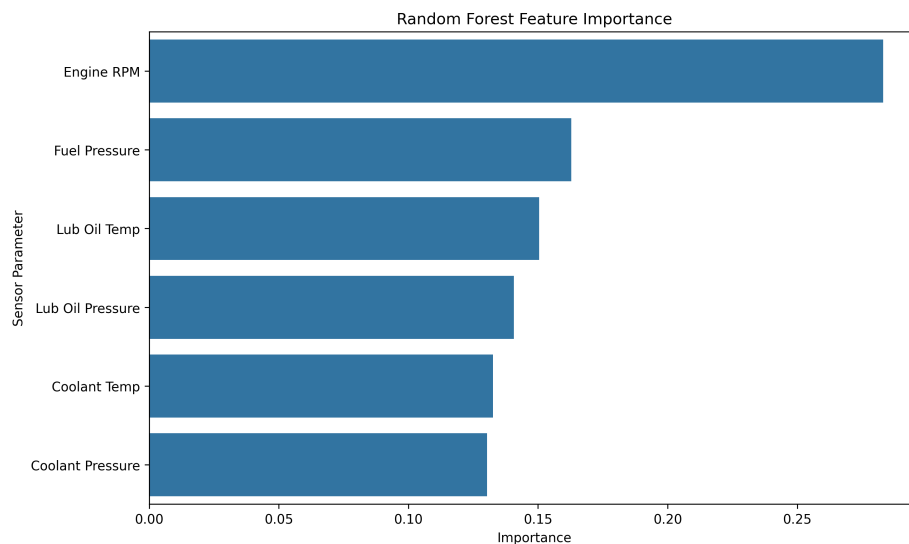


Figure 5.1: Random Forest Feature Importance

The feature importance analysis provides an indication of the relative contribution of the sensor parameters to the model.

5.2 XGBoost Feature Importance

Feature importance was also calculated for the XGBoost model.

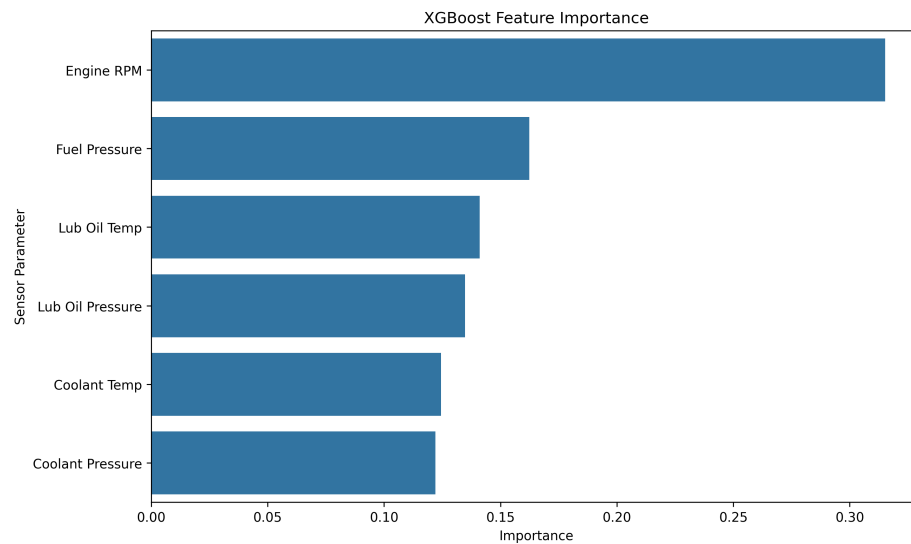


Figure 5.2: XGBoost Feature Importance

5.3 SHAP Explainability

SHAP was used with the tree-based model to provide a more detailed explanation of feature contributions.

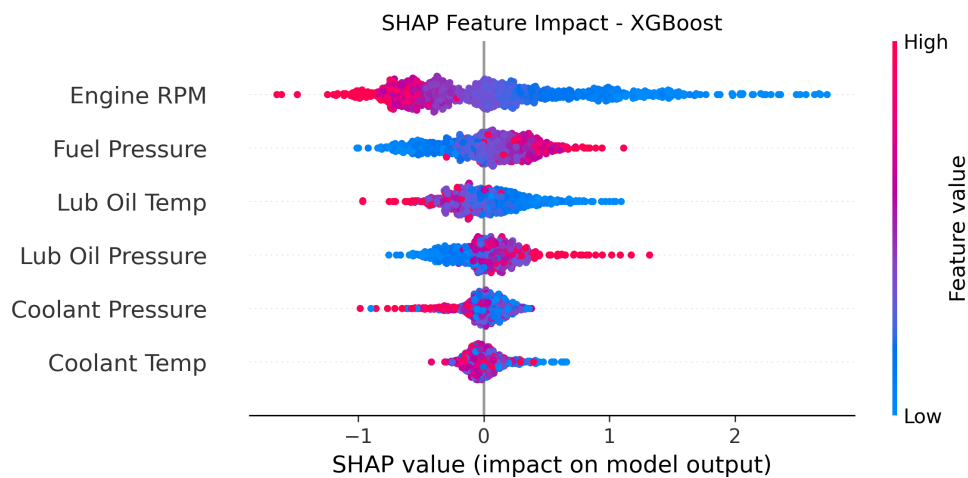


Figure 5.3: SHAP Summary Plot

The SHAP summary plot provides information about both feature importance and the direction of feature contributions across the dataset.

5.4 SHAP Feature Importance

The mean absolute SHAP values obtained from the model are shown in Table 5.1.

Table 5.1: SHAP Feature Importance

Feature	Mean Absolute SHAP Value
Engine RPM	0.5803694
Fuel Pressure	0.2714020
Lub Oil Temp	0.2129758
Lub Oil Pressure	0.1888995
Coolant Pressure	0.1180625
Coolant Temp	0.0942661

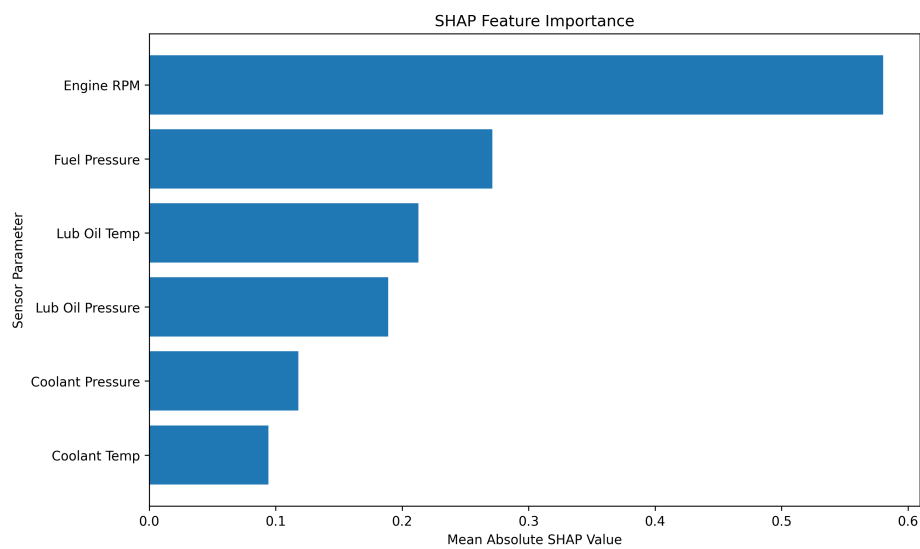


Figure 5.4: SHAP-Based Feature Importance

The results indicate that Engine RPM has the highest mean absolute SHAP value among the features used in the model.

Chapter 6

Vehicle Health Monitoring

6.1 Failure Probability

The selected machine learning model provides the probability that an observation belongs to the maintenance-required class.

Let the predicted failure probability be:

$$P_f = P(Y = 1|X) \quad (6.1)$$

where:

- P_f represents failure probability.
- $Y = 1$ represents the maintenance-required condition.
- X represents the observed sensor values.

6.2 Vehicle Health Score

The failure probability is converted into a health score ranging from 0 to 100.

The formula used is:

$$Health\ Score = 100 \times (1 - P_f) \quad (6.2)$$

Therefore:

- Higher failure probability produces a lower health score.
- Lower failure probability produces a higher health score.

6.3 Health Status Categories

The system categorises the predicted condition using the following thresholds.

Table 6.1: Vehicle Health Status Thresholds

Failure Probability	Status
$P_f < 0.25$	NORMAL
$0.25 \leq P_f < 0.50$	MONITOR
$0.50 \leq P_f < 0.75$	WARNING
$P_f \geq 0.75$	CRITICAL

6.4 Vehicle Health Score Visualisation

The generated health-score visualisation demonstrates how the predicted probability can be transformed into an interpretable vehicle health measure.

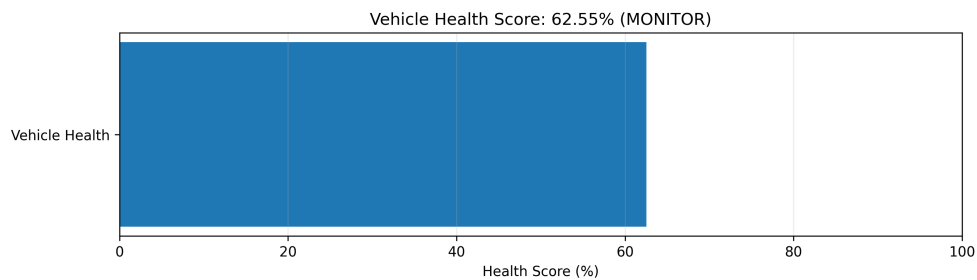


Figure 6.1: Vehicle Health Score

6.5 Example Prediction

One representative sample prediction produced the following values:

Table 6.2: Representative Vehicle Health Prediction

Parameter	Value
Failure Probability	0.6189313
Health Score	38.1069
Status	WARNING

This example demonstrates the conversion of a model probability into a health score and maintenance status.

Chapter 7

Experimental Setup

7.1 Software Environment

The project was implemented using Python and executed in a notebook-based environment.

The primary libraries used were:

Table 7.1: Software Libraries Used

Library	Purpose
Python	Programming language
Pandas	Data manipulation
NumPy	Numerical computation
Scikit-learn	Machine learning and evaluation
XGBoost	Gradient boosting classifier
SHAP	Model explainability
Matplotlib	Data visualisation
Seaborn	Statistical visualisation
Joblib	Model serialisation

7.2 Machine Learning Models

Two machine learning models were implemented:

1. Random Forest Classifier
2. XGBoost Classifier

Both models were trained using the same six sensor parameters.

7.3 Evaluation Metrics

7.3.1 Accuracy

Accuracy represents the proportion of correctly classified observations.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7.1)$$

7.3.2 Precision

Precision represents the proportion of predicted positive observations that are actually positive.

$$Precision = \frac{TP}{TP + FP} \quad (7.2)$$

7.3.3 Recall

Recall represents the proportion of actual positive observations correctly identified by the model.

$$Recall = \frac{TP}{TP + FN} \quad (7.3)$$

7.3.4 F1 Score

The F1 Score combines precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (7.4)$$

7.3.5 ROC-AUC

ROC-AUC measures the model's ability to distinguish between the two classes across classification thresholds.

Chapter 8

Results and Analysis

8.1 Model Performance

The obtained evaluation results are shown in Table 8.1.

Table 8.1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	65.11%	73.48%	69.87%	71.63%	68.83%
XGBoost	65.88%	69.67%	81.24%	75.01%	68.60%

8.2 Model Selection

The project uses ROC-AUC as the criterion for selecting the model for the vehicle health monitoring stage.

The Random Forest model obtained a ROC-AUC of approximately 68.83%, while XGBoost obtained approximately 68.60%.

Therefore, based on the specified ROC-AUC selection criterion, the Random Forest model was used for the final vehicle health monitoring pipeline.

It is important to note that XGBoost obtained higher recall and F1 Score in the evaluated results, while Random Forest obtained the slightly higher ROC-AUC.

8.3 Feature Importance Results

The Random Forest and XGBoost feature importance results show that Engine RPM and Fuel Pressure are among the higher-ranked features.

The SHAP analysis similarly indicates a comparatively high contribution from Engine RPM.

8.4 XGBoost Feature Importance Values

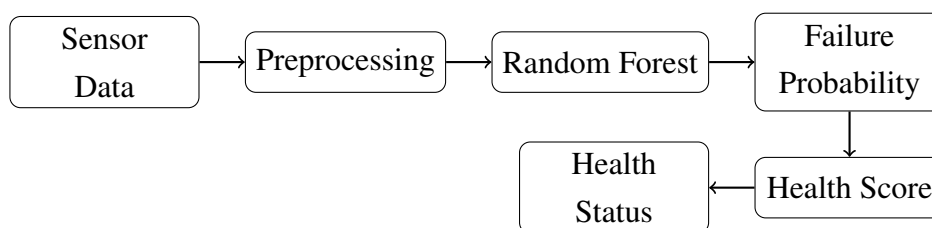
The feature importance values obtained from XGBoost are shown below.

Table 8.2: XGBoost Feature Importance

Feature	Importance
Engine RPM	0.3152351
Fuel Pressure	0.1624154
Lub Oil Temp	0.1410243
Lub Oil Pressure	0.1348149
Coolant Temp	0.1245126
Coolant Pressure	0.1219976

8.5 Overall Result

The completed system successfully performs the following pipeline:



The resulting system demonstrates a complete predictive maintenance workflow from raw sensor measurements to an interpretable vehicle health status.

Chapter 9

Discussion

9.1 Model Behaviour

Both Random Forest and XGBoost were able to learn predictive patterns from the engine sensor measurements.

The two models produced different metric profiles. XGBoost produced higher recall and F1 Score, whereas Random Forest produced a slightly higher ROC-AUC. Since ROC-AUC was selected as the model-selection criterion for this project, Random Forest was used in the final health monitoring pipeline.

9.2 Importance of Engine RPM

Engine RPM obtained the highest SHAP importance among the six features.

This indicates that variations in Engine RPM contributed substantially to the model's predictions within the evaluated dataset.

However, feature importance should be interpreted as a model-specific result and does not by itself establish a causal relationship between a sensor parameter and mechanical failure.

9.3 Explainability

The use of SHAP improves transparency by allowing the contribution of individual sensor features to be examined.

This is particularly useful in predictive maintenance applications because maintenance personnel may need information about why a prediction was generated rather than only receiving a binary output.

9.4 Health Monitoring

The conversion from failure probability to a 0–100 health score provides an easier-to-understand representation of model output.

The four status categories provide an additional layer of interpretation:

- NORMAL
- MONITOR
- WARNING
- CRITICAL

Chapter 10

Limitations

The current implementation has several limitations.

1. The model is trained using six sensor parameters available in the selected dataset.
2. Parameters such as vibration, battery voltage, vehicle speed, and operating hours were not available in the selected dataset and therefore were not artificially included.
3. The ROC-AUC values obtained by the models are moderate, indicating that the current implementation has room for improvement.
4. The health-score thresholds are predefined rules and are not dynamically learned from maintenance outcomes.
5. The system is based on a static tabular dataset and does not currently process continuous real-time sensor streams.
6. The system does not directly interface with a physical vehicle ECU or CAN bus.
7. The model's predictions depend on the quality and representativeness of the training dataset.
8. A machine learning prediction should be treated as decision-support information rather than a replacement for professional mechanical inspection.

Chapter 11

Future Scope

Several improvements can be considered for future versions of the project.

11.1 Real-Time Sensor Integration

The system can be connected to real-time vehicle sensor streams through suitable automotive interfaces such as CAN-based data acquisition systems.

11.2 Additional Sensor Parameters

Future datasets could incorporate additional parameters such as:

- Vibration
- Battery voltage
- Vehicle speed
- Operating hours
- Engine load
- Exhaust temperature
- Additional pressure measurements

11.3 Deep Learning

Sequential sensor data could be analysed using:

- LSTM

- GRU
- Transformer-based models

These approaches may be useful when temporal relationships between sensor readings are available.

11.4 Remaining Useful Life Prediction

Instead of only predicting whether maintenance is required, a future system could estimate Remaining Useful Life (RUL) for specific components.

11.5 Real-Time Dashboard

A web-based dashboard could display:

- Current sensor readings
- Failure probability
- Vehicle health score
- Current maintenance status
- Feature importance
- SHAP explanations
- Historical health trends

11.6 Automated Maintenance Alerts

The system could be extended to send maintenance notifications when the health score or failure probability crosses a predefined threshold.

Chapter 12

Conclusion

This project developed an AI/ML-based Predictive Maintenance and Vehicle Health Monitoring System using real engine sensor data.

The system processes six sensor parameters consisting of Engine RPM, Lub Oil Pressure, Fuel Pressure, Coolant Pressure, Lub Oil Temperature, and Coolant Temperature. Random Forest and XGBoost classifiers were implemented and evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

The Random Forest model achieved a ROC-AUC of approximately 68.83%, while XGBoost achieved approximately 68.60%. Since ROC-AUC was used as the model-selection criterion, Random Forest was selected for the final vehicle health monitoring pipeline.

SHAP was incorporated to improve model interpretability. The analysis showed that Engine RPM had the highest mean absolute SHAP contribution among the evaluated sensor parameters.

The final system converts failure probability into a vehicle health score using:

$$Health\ Score = 100 \times (1 - P_f) \quad (12.1)$$

The resulting score is then mapped to NORMAL, MONITOR, WARNING, or CRITICAL maintenance states.

Overall, the project demonstrates a complete machine learning pipeline for predictive engine condition classification and interpretable vehicle health monitoring. The system also provides a foundation for future development involving real-time sensor integration, additional automotive parameters, deep learning, remaining useful life prediction, and deployment through a real-time monitoring dashboard.

Bibliography

- [1] Scikit-learn Documentation, *Machine Learning in Python*, <https://scikit-learn.org/>
- [2] XGBoost Documentation, *XGBoost: An Optimized Gradient Boosting Library*, <https://xgboost.readthedocs.io/>
- [3] SHAP Documentation, *SHAP: Explainable AI*, <https://shap.readthedocs.io/>
- [4] Hugging Face Datasets, *Predictive Maintenance Dataset*, <https://huggingface.co/datasets/jskswamy/predictive-maintenance-data>
- [5] Pandas Documentation, *Python Data Analysis Library*, <https://pandas.pydata.org/>
- [6] NumPy Documentation, *Scientific Computing with Python*, <https://numpy.org/>
- [7] Matplotlib Documentation, *Visualization with Python*, <https://matplotlib.org/>

Appendix A

Project Directory Structure

The final GitHub project contains the following major components:

```
Module-3-Predictive-Maintenance/
|
|-- graphs/
|   |-- 01_engine_condition_distribution.png
|   |-- 02_sensor_distributions.png
|   |-- 03_correlation_matrix.png
|   |-- 04_model_comparison.png
|   |-- 05_roc_curve.png
|   |-- 06_confusion_matrix.png
|   |-- 07_random_forest_feature_importance.png
|   |-- 08_xgboost_feature_importance.png
|   |-- 09_shap_summary.png
|   |-- 10_shap_feature_importance.png
|   |-- 11_vehicle_health_score.png
|
|-- models/
|   |-- random_forest_model.pkl
|   |-- xgboost_model.pkl
|
|-- results/
|   |-- cleaned_engine_sensor_dataset.csv
|   |-- model_comparison.csv
|   |-- project_metadata.json
|   |-- sample_predictions.csv
|   |-- shap_feature_importance.csv
|   |-- xgboost_feature_importance.csv
```

```
|  
|-- Module_3_Predictive_Maintenance_Vehicle_Health_Monitoring.ipynb  
|-- PROJECT_SUMMARY.txt  
|-- README.md  
|-- requirements.txt  
'-- .gitattributes
```

Appendix B

Health Score Calculation

The health monitoring system first obtains the probability of the maintenance-required class:

$$P_f = P(Y = 1|X) \quad (\text{B.1})$$

The health score is then calculated as:

$$H = 100(1 - P_f) \quad (\text{B.2})$$

For example, for:

$$P_f = 0.6189313 \quad (\text{B.3})$$

the resulting score is approximately:

$$H = 100(1 - 0.6189313) \quad (\text{B.4})$$

$$H \approx 38.1069 \quad (\text{B.5})$$

The resulting status falls within the WARNING category because the failure probability lies between 0.50 and 0.75.

Appendix C

Repository Information

The complete project source code, notebook, trained models, result files, graphs, and documentation are maintained in the GitHub repository:

<https://github.com/ParthibSarbjna/Module-3-Predictive-Maintenance>

The repository contains the complete implementation of the predictive maintenance and vehicle health monitoring pipeline.